Experiment 11

Title

Pneumonia Diagnosis using Machine Learning on Chest X-Ray Derived Features

SAP ID:	590013857
Dataset:	`https://www.kaggle.com/datasets/aadigupta1601/chest-x-ray-pneumonia-numeri`

AIM

To develop a machine learning-based system for pneumonia detection using numerical features extracted from chest X-ray images and evaluate different models.

THEORY

Pneumonia detection using chest X-rays can be automated using machine learning techniques. Instead of raw images, this experiment uses pre-extracted numerical features inspired by radiomics.

Key feature groups:

- **Global Features**: Mean, variance, percentiles
- **Texture Features (GLCM)**: Contrast, homogeneity
- **Frequency Features (FFT)**: Signal patterns
- **Edge Features**: Edge density and strength
- **LBP Features**: Local texture patterns

Machine learning models used:

- Logistic Regression
- SVM (RBF)
- KNN
- Random Forest
- Gradient Boosting

Dimensionality reduction:

- PCA (Principal Component Analysis)
- Mutual Information Feature Selection

Commands/Code Used

```
# Load dataset
train_df = load_split("train_features.csv")

# Feature Engineering
df["intensity_range"] = df["max"] - df["min"]

# Models
models = [
    LogisticRegression(),
    SVM(),
    KNN(),
    RandomForest(),
    GradientBoosting()
]

# PCA
pca = PCA(n_components=0.95)

# Evaluation
accuracy, precision, recall, f1
```

Results

Feature Group Analysis

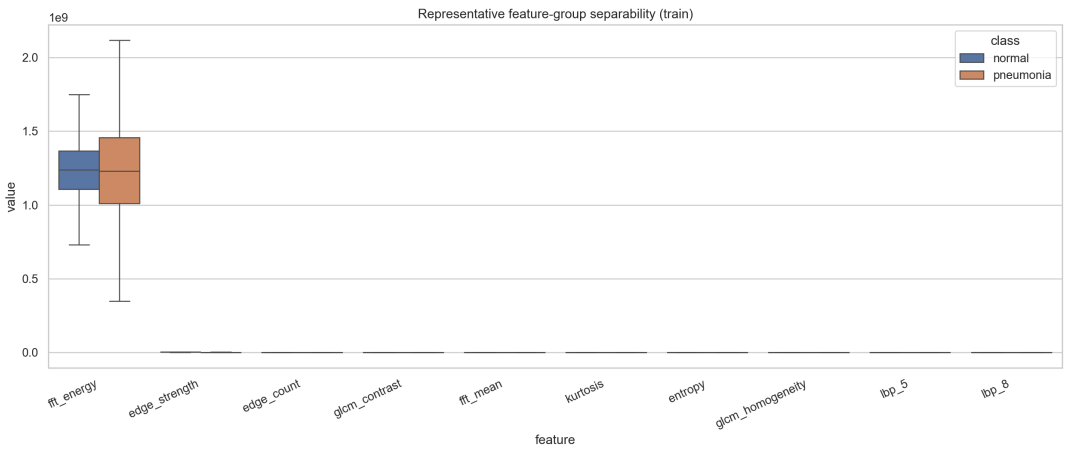

Figure 1: Feature group separability between normal and pneumonia

PCA Explained Variance

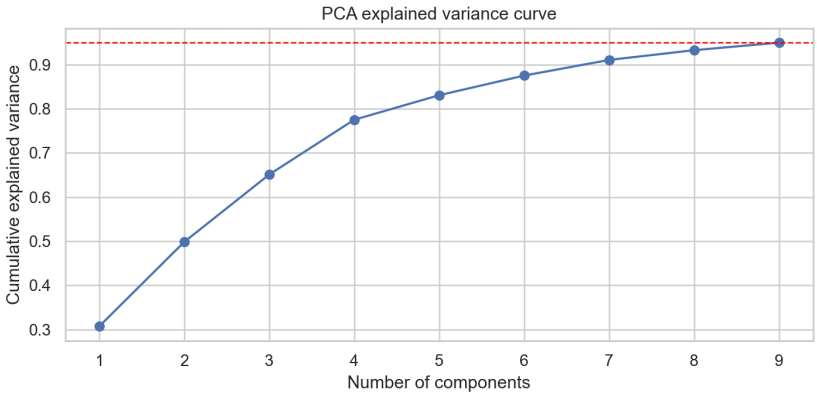

Figure 2: PCA cumulative variance curve

Class Distribution

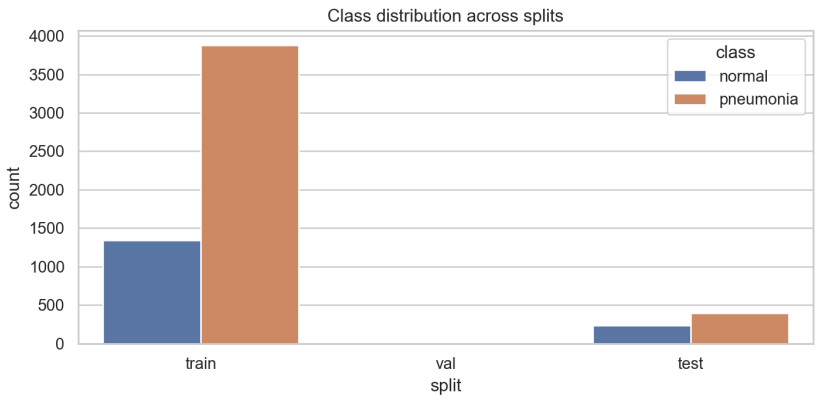

Figure 3: Dataset class imbalance

Correlation Heatmap

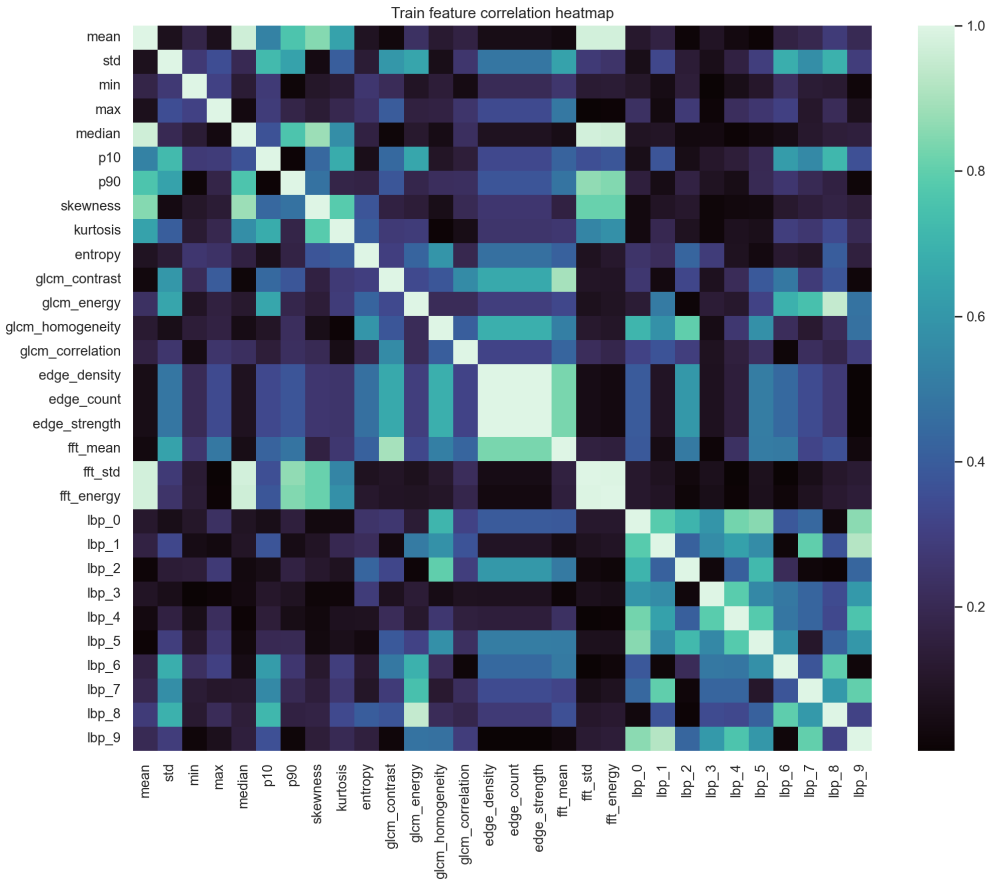

Figure 4: Feature correlation heatmap

Confusion Matrix (Best Model)

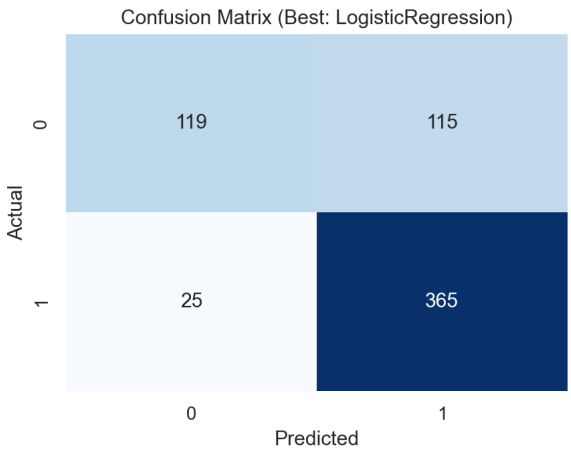

Figure 5: Confusion matrix of best model

Model Performance

Best model (validation): **Logistic Regression** :contentReference[oaicite:0]index=0

Test Results:

- Accuracy: 0.7756
- Precision: 0.7852
- Recall: 0.7756
- F1-Score: 0.7605

Detailed performance:

- Pneumonia class has high recall (93.5%)
- Normal class has lower recall (50.8%)

These values are obtained from the final classification report :contentReference[oaicite:1]index=1

Conclusion

In this experiment, a complete machine learning pipeline was developed for pneumonia detection using chest X-ray derived numerical features.

Feature engineering and dimensionality reduction techniques such as PCA improved model efficiency. Among all models, Logistic Regression performed best based on validation F1-score, while SVM showed strong test performance.

The model achieved high recall for pneumonia detection, which is critical in medical diagnosis to minimize false negatives.

This experiment demonstrates how machine learning can assist in medical imaging and decision-making systems.